

PrimeCore Minerals

Firewall Controls

Implementation

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1. Introduction

This document is a project artefact that explains how the firewall controls were designed, configured and tested across the PrimeCore Minerals Brisbane HQ and Mine Site A networks. Both locations use pfSense as the main firewall and gateway for their internal VLANs, allowing traffic between the different networks to be controlled through firewall rules.

The firewall rules were designed around least privilege, meaning users and devices are only given access to the networks and services they need. For example, visitor devices can still access the required services while being blocked from the internal PrimeCore Minerals networks. Additional firewall controls are also used to help protect the management, IoT, CCTV, backup and server networks.

2. Purpose and Scope

The firewall controls were implemented to keep the different parts of the Brisbane HQ and Mine Site A networks separated while still allowing employees, devices and services to communicate where needed. This supports the project MVP by applying network segmentation, access control and logging across both locations. The pfSense firewalls at each site also provide a security boundary between the Brisbane HQ and Mine Site A. This allows traffic travelling between the two locations to be controlled and securely transferred through the site-to-site VPN.

- Separate employee, visitor, printer, video, CCTV, IoT, server, backup and management traffic using VLAN-based firewall boundaries
- Allow required services such as DNS, HTTP/HTTPS, printing, monitoring and authentication without providing unnecessary network access
- Protect management interfaces from normal employee and visitor devices.
- Keep visitor and contractor devices isolated from PrimeCore Minerals internal networks.
- Log important denied traffic and forward security events to the central monitoring platform.
- Control approved traffic entering and leaving the IKEv2/IPsec site-to-site VPN.

3. Firewall Architecture and Aliases

To make the firewall rules easier to set up and manage, I created aliases in pfSense for the different networks. This makes creating firewall rules easier because I can use the network name instead of entering the subnet each time.

3.1 Network Aliases

Brisbane HQ

Alias	Network	Purpose
EMPLOYEE_LAN	192.168.10.0/24	Wired employee device
EMPLOYEE_WIFI	192.168.11.0/24	Employee wireless devices
VISITOR_NETWORK	192.168.20.0/24	Visitor and contractor devices
PRINTER_NETWORK	192.168.30.0/24	Network printers
VIDEO_NETWORK	192.168.40.0/24	Video conferencing devices
CCTV_NETWORK	192.168.41.0/24	CCTV devices
IoT_NETWORK	192.168.50.0/24	IoT devices
SERVER_NETWORK	192.168.90.0/24	Servers and monitoring systems
BACKUP_NETWORK	192.168.91.0/24	Backup systems
MANAGEMENT_NETWORK	192.168.99.0/24	Network management devices
INTERNAL_NETWORKS	Multiple subnets	Groups the internal PrimeCore networks for firewall rules

The INTERNAL_NETWORKS alias groups the internal PrimeCore networks together so they can be referenced within a single firewall rule. The Visitor network is not included because Visitor devices should remain separated from the internal network.

Mine-Site A

Alias	Network	Purpose
SITEA_EMPLOYEE	192.168.110.0/24	Wired employee device
SITEA_EMP_WIFI	192.168.111.0/24	Employee wireless devices
SITEA_VISITOR	192.168.120.0/24	Visitor and contractor devices
SITEA_PRINTER	192.168.130.0/24	Network printers
SITEA_VIDEO	192.168.140.0/24	Video conferencing devices
SITEA_CCTV	192.168.141.0/24	CCTV devices
SITEA_IOT	192.168.150.0/24	IoT devices
SITEA_SERVER	192.168.190.0/24	Servers and monitoring systems
SITEA_BACKUP	192.168.191.0/24	Backup systems
SITEA_MANAGEMENT	192.168.199.0/24	Network management devices
SITEA_INTERNAL_NETWORKS	Multiple subnets	Groups the internal PrimeCore networks for firewall rules

The Site A VLANs use the same VLAN IDs as the Brisbane HQ, but each one uses its own 192.168.1xx.0/24 network. Keeping the VLAN layout consistent between both locations while making sure each subnet is different for routing and the site-to-site VPN.

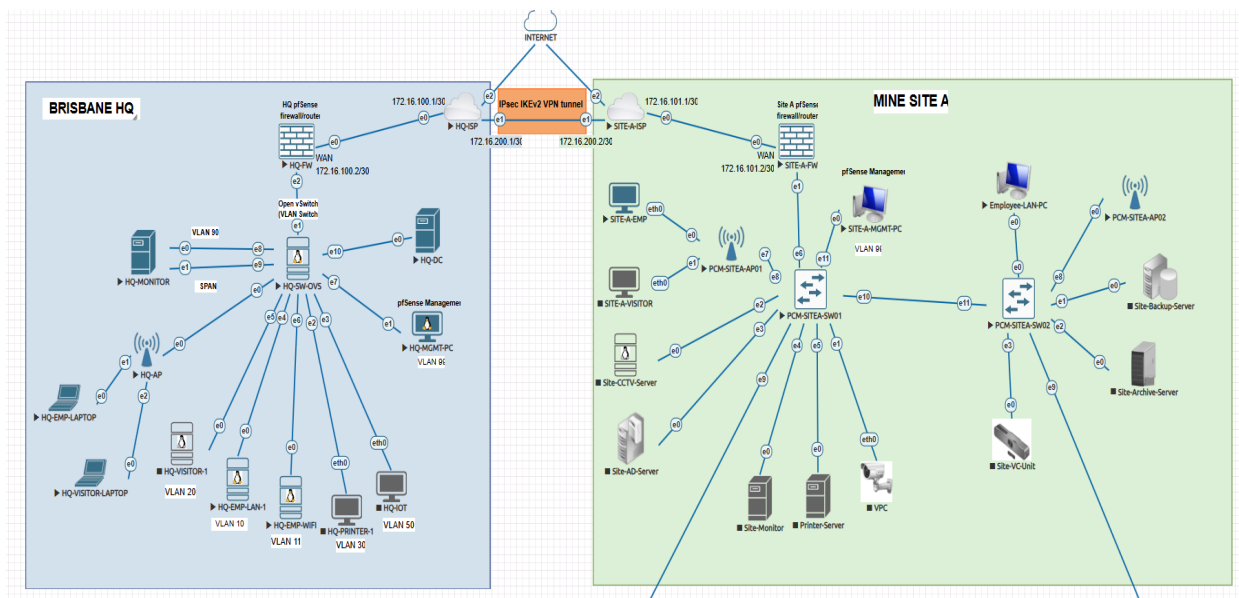


Figure 1: PrimeCore Minerals network topology showing Brisbane HQ and Mine Site A

3.2 Port Aliases

Alias	Ports	Purpose
WEB_PORTS	TCP 80, 443	HTTP and HTTPS web traffic
DNS_PORT	TCP/UDP 53	DNS traffic
MANAGEMENT_PORTS	TCP 22, 443	SSH and HTTPS management access
PRINT_PORTS	TCP 631, 9100, 515	Network printing services
NTP_PORT	UDP 123	Network time synchronisation
IPSEC_PORTS	500, 4500	IKE/IPsec VPN negotiation
AD_TCP_PORTS	53, 88, 135, 389, 445, 464, 636, 3268, 3269	Active Directory TCP services
AD_UDP_PORTS	53, 88, 123, 389, 464	Active Directory UDP services

Port aliases were also created for services that are used across multiple firewall rules. Which makes the rules easier to read, manage, and update. They also allow access to be limited to the specific services required. For example, Employee devices are allowed to access the Printer VLAN using the required printing ports rather than being given unrestricted access to the entire network.

3.3 Host Aliases

Alias	Address	Purpose
AD_SERVER	192.168.90.10	Brisbane HQ Active Directory server
HQ_AP	192.168.99.61	Brisbane HQ wireless access point

Firewall Aliases IP				
Name	Type	Values	Description	Actions
BACKUP_NETWORK	Network(s)	192.168.91.0/24		
CCTV_NETWORK	Network(s)	192.168.41.0/24		
EMPLOYEE_LAN	Network(s)	192.168.10.0/24		
EMPLOYEE_WIFI	Network(s)	192.168.11.0/24		
INTERNAL_NETWORKS	Network(s)	192.168.10.0/24, 192.168.11.0/24, 192.168.30.0/24, 192.168.40.0/24, 192.168.41.0/24, 192.168.50.0/24, 192.168.90.0/24, 192.168.91.0/24, 192.168.99.0/24		
IOT_NETWORK	Network(s)	192.168.50.0/24		
MANAGEMENT_NETWORK	Network(s)	192.168.99.0/24		
PRINTER_NETWORK	Network(s)	192.168.30.0/24		
SERVER_NETWORK	Network(s)	192.168.90.0/24		
VIDEO_NETWORK	Network(s)	192.168.40.0/24		
VISITOR_NETWORK	Network(s)	192.168.20.0/24		

Figure 2: Brisbane HQ pfSense network aliases, including the *INTERNAL_NETWORKS* alias used by segmentation rules.

Once the network and port aliases were created, they were used to implement the firewall rules for each of the established VLANs. Each rule was designed and implemented based on the requirements of each VLAN and who they should be accessed by. Allow rules were added when a service such as DNS, web access, and printer was required, while block rules were used to stop unnecessary communication between restricted networks.

For machines that need additional restricted access, specific firewall rules were created. For example, HQ-MGMT-PC is only allowed to reach the Grafana dashboard on HQ-MONITOR using TCP port 3000, while HQ-MONITOR was given the DNS and web access required for logging and monitoring software.

4. Brisbane HQ

4.1 Employee VLAN Controls

The Employee VLAN provides staff with normal network access while preventing direct access to the Management VLAN. Separate rules allow web browsing, DNS, printing, Active Directory access and the specific Site A VPN test path.

Control	Source	Destination	Service	Action
R05 Management restriction	EMPLOYEE_VLAN	MANAGEMENT_NETWORK	Any	Block + Log
Site A Employee VPN access	EMPLOYEE_VLAN	192.168.110.0/24	ICMP	Allow
R02a Employee web access	EMPLOYEE_VLAN	Any	WEB_PORTS	Allow
R09a Employee printing	EMPLOYEE_VLAN	PRINTER_NETWORK	PRINT_PORTS	Allow
R02b Employee DNS	EMPLOYEE_VLAN	Any	DNS_PORT	Allow
Active Directory TCP	EMPLOYEE_LAN	AD_SERVER	AD_TCP_PORTS	Allow
Active Directory UDP	EMPLOYEE_LAN	AD_SERVER	AD_UDP_PORTS	Allow

4.2 Employee Wi-Fi VLAN Controls

The Employee Wi-Fi network is given similar access to the wired Employee VLAN. It can use web, DNS, printing and Active Directory services, while access to the Management VLAN is blocked.

Control	Source	Destination	Service	Action
R06 Management restriction	EMPLOYEE_WIFI_VLAN	MANAGEMENT_NETWORK	Any	Block + Log
R03a Employee Wi-Fi web	EMPLOYEE_WIFI_VLAN	Any	WEB_PORTS	Allow
R09b Employee Wi-Fi printing	EMPLOYEE_WIFI_VLAN	PRINTER_NETWORK	PRINT_PORTS	Allow

R03b Employee Wi-Fi DNS	EMPLOYEE_WIFI_VLAN	Any	DNS_PORT	Allow
Active Directory TCP	EMPLOYEE_WIFI	AD_SERVER	AD_TCP_PORTS	Allow
Active Directory UDP	EMPLOYEE_WIFI	AD_SERVER	AD_UDP_PORTS	Allow

4.3 Visitor VLAN Controls

The Visitor VLAN is kept separate from the internal company networks. Visitors can still use DNS and access normal web services, but they are blocked from reaching employee, server, management and other internal VLANs.

Control	Source	Destination	Service	Action
Visitor laptop logging	192.168.20.101	192.168.90.60	UDP 514	Allow
Visitor AD SMB	192.168.20.101	192.168.90.10	TCP 445	Allow
Visitor AD Global Catalog	192.168.20.101	192.168.90.10	TCP 3268	Allow
Visitor AD NTP	VISITOR_VLAN	192.168.90.10	UDP 123	Allow
Visitor AD authentication services	VISITOR_VLAN	192.168.90.10	AD UDP/TCP services	Allow
R04 Visitor internal isolation	VISITOR_VLAN	INTERNAL_NETWORKS	Any	Block + Log
R01a Visitor web access	VISITOR_VLAN	Any	WEB_PORTS	Allow
R01b Visitor DNS	VISITOR_VLAN	Any	DNS_PORT	Allow

4.4 Printer VLAN Controls

Employee and Employee Wi-Fi devices already have specific PRINT_PORTS rules that allow them to send print traffic to VLAN 30. The following rules are so printers cannot initiate unrestricted connections back into the internal network.

Control	Source	Destination	Service	Action
Printer DNS	PRINTER_VLAN	192.168.30.1	TCP/UDP 53	Allow
Printer gateway test	PRINTER_VLAN	192.168.30.1	ICMP	Allow
Printer internal restriction	PRINTER_VLAN	INTERNAL_NETWORKS	Any	Block + Log
Printer Visitor separation	PRINTER_VLAN	VISITOR_VLAN	Any	Block + Log

Printer web/update access	PRINTER_VLAN	Any	WEB_PORTS	Allow
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4.5 Video VLAN Controls

These controls are used to keep the video-conferencing equipment separate from the rest of the HQ network, while still allowing access to the basic services it needs to operate.

Control	Source	Destination	Service	Action
Video DNS	VIDEO_VLAN	192.168.40.1	TCP/UDP 53	Allow
Video gateway test	VIDEO_VLAN	192.168.40.1	ICMP	Allow
Video internal restriction	VIDEO_VLAN	INTERNAL_NETWORKS	Any	Block + Log
Video Visitor separation	VIDEO_VLAN	VISITOR_VLAN	Any	Block + Log
Video web access	VIDEO_VLAN	Any	WEB_PORTS	Allow

4.6 CCTV VLAN Controls

The CCTV VLAN is kept as a restricted network. CCTV devices are blocked from starting connections to the internal company networks, and any blocked traffic is logged so it can be reviewed.

Control	Source	Destination	Service	Action
R12 Unauthorised CCTV internal access	CCTV_VLAN	INTERNAL_NETWORKS	Any	Block + Log

4.7 IoT VLAN Controls

The IoT VLAN is kept separate from the more important internal networks. This helps stop IoT devices from being able to directly access employee, server or management systems if one of them was ever compromised.

Control	Source	Destination	Service	Action
R011 Block IoT access to internal network	IOT_VLAN	INTERNAL_NETWORKS	Any	Block + Log

4.8 Server VLAN Controls

The Server VLAN contains HQ infrastructure, including HQ-DC and HQ-MONITOR. The current rules allow the specific outbound services required by these servers rather than using one unrestricted Server VLAN allow rule.

Control	Source	Destination	Service	Action
HQ-MONITOR Gmail SMTP	192.168.90.60	Any	TCP 587	Allow
HQ-DC web access	AD_SERVER	Any	WEB_PORTS	Allow
HQ-DC DNS	AD_SERVER	Any	DNS_PORT	Allow
HQ-DC NTP	AD_SERVER	Any	NTP_PORT	Allow
HQ-MONITOR DNS	192.168.90.60	Any	DNS_PORT	Allow
HQ-MONITOR web	192.168.90.60	Any	WEB_PORTS	Allow

4.9 Backup VLAN Controls

Control	Source	Destination	Service	Action
Backup internal restriction	BACKUP_VLAN	INTERNAL_NETWORKS	Any	Block + Log

4.10 Management VLAN Controls

The Management VLAN is used for administrative access to pfSense and the monitoring dashboard. The current rules also allow the HQ access point to use the services it needs.

Control	Source	Destination	Service	Action
R10b Management to pfSense	MANAGEMENT_VLAN	This Firewall	MANAGEMENT_PORTS	Allow
R10a Management to pfSense	MANAGEMENT_VLAN	This Firewall	MANAGEMENT_PORTS	Allow
HQ-AP DNS	HQ_AP	Any	DNS_PORT	Allow
HQ-AP web access	HQ_AP	Any	WEB_PORTS	Allow
HQ-AP NTP	HQ_AP	Any	NTP_PORT	Allow
Grafana dashboard	192.168.99.71	192.168.90.60	TCP 3000	Allow

4.11 IPsec Interface Controls

The IPsec rules control what traffic is allowed through the VPN once it reaches HQ-FW. The Phase 2 settings choose which HQ and Site A networks can use the tunnel, while the firewall rules decide what traffic between those networks is actually allowed.

Control	Source	Destination	Service	Action
Site A Employee to HQ Employee	192.168.110.0/24	192.168.10.0/24	ICMP	Allow
Site A central firewall logging	192.168.190.0/24	192.168.90.60	UDP 514	Allow
Site A Server to HQ Server validation	192.168.190.0/24	192.168.90.0/24	ICMP	Allow

Rules (Drag to Change Order)											
☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	✓ 2/76 KIB	IPv4 TCP	MANAGEMENT_VLAN subnets	*	This Firewall (self)	MANAGEMENT_PORTS	*	none		R10b -	
☐	✓ 0/0 B	IPv4 TCP	MANAGEMENT_VLAN subnets	*	This Firewall (self)	MANAGEMENT_PORTS	*	none		R10a - Allow Management Access to pfSense	
☐	✓ 0/16 KIB	IPv4 TCP/UDP	HQ_AP	*	*	DNS_PORT	*	none		HQ AP DNS	
☐	✓ 0/2.00 MIB	IPv4 TCP	HQ_AP	*	*	WEB_PORTS	*	none			
☐	✓ 0/0 B	IPv4 UDP	*	*	*	NTP_PORT	*	none			
☐	✓ 0/0 B	IPv4 TCP	192.168.99.71	*	192.168.90.60	3000 (HBCI)	*	none			

Figure 3: Brisbane HQ Management VLAN firewall rules showing pfSense, access point and Grafana access controls.

5. Mine Site A

Mine Site A was configured with the same segmentation intent as HQ but adapted to the Site A addressing scheme. The rule set was progressively tested using the actual Site A endpoints and VPCS devices. Gateway ICMP rules were intentionally limited to each VLAN's own pfSense interface address so that a device could not use a broad "This Firewall" rule to reach another VLAN gateway.

5.1 Employee and Employee Wi-Fi

Control	Source	Destination	Service	Action
Management restriction	SITEA_EMPLOYEE 192.168.110.0/24	192.168.199.0/24	Any	Block + Log
DNS	SITEA_EMPLOYEE	SITE-A-FW	TCP/UDP 53	Allow
Printer access	SITEA_EMPLOYEE	192.168.130.0/24	PRINT_PORTS	Allow
Gateway ICMP	SITEA_EMPLOYEE	192.168.110.1	ICMP	Allow
Web access	SITEA_EMPLOYEE	Any	TCP 80/443	Allow
HQ employee VPN test	SITEA_EMPLOYEE	192.168.10.0/24	ICMP	Allow
Management restriction	SITEA_EMP_WIFI 192.168.111.0/24	192.168.199.0/24	Any	Block + Log
DNS / print / web	SITEA_EMP_WIFI	Required destinations	DNS, PRINT_PORTS, WEB_PORTS	Allow

5.2 Visitor VLAN controls

The Site A Visitor VLAN was tested using the SITE-A-VISITOR VPCS. It received an IP address through DHCP, could reach its own gateway and was blocked from accessing the internal Site A networks.

Control	Source	Destination	Service	Action
DNS	SITEA_VISITOR	192.168.120.1	TCP/UDP 53	Allow
Gateway test	SITEA_VISITOR	192.168.120.1	ICMP	Allow
Internal isolation	SITEA_VISITOR	SITEA_INTERNAL_NETWORKS	Any	Block + Log
Web access	SITEA_VISITOR	Any	TCP 80/443	Allow

5.3 CCTV VLAN Controls

The Site A CCTV VLAN is kept separate from the main internal networks. The firewall allows the basic services the CCTV devices need, while access to the Employee, Management and other internal networks is blocked and logged.

Control	Source	Destination	Service	Action
DNS	SITEA_CCTV	192.168.141.1	TCP/UDP 53	Allow
Gateway ICMP	SITEA_CCTV	192.168.141.1	ICMP	Allow
Internal restriction	SITEA_CCTV	SITEA_INTERNAL_NETWORKS	Any	Block + Log
Visitor separation	SITEA_CCTV	SITEA_VISITOR	Any	Block + Log

5.4 Video VLAN Controls

Control	Source	Destination	Service	Action
DNS	SITEA_VIDEO	192.168.140.1	TCP/UDP 53	Allow
Gateway ICMP	SITEA_VIDEO	192.168.140.1	ICMP	Allow
Internal restriction	SITEA_VIDEO	SITEA_INTERNAL_NETWORKS	Any	Block + Log
Visitor separation	SITEA_VIDEO	SITEA_VISITOR	Any	Block + Log
Web access	SITEA_VIDEO	Any	TCP 80/443	Allow

5.5 Backup VLAN Controls

Control	Source	Destination	Service	Action
DNS	SITEA_BACKUP	192.168.191.1	TCP/UDP 53	Allow
Gateway ICMP	SITEA_BACKUP	192.168.191.1	ICMP	Allow
Backup to Server	SITEA_BACKUP	SITEA_SERVER 192.168.190.0/24	Required backup traffic / prototype Any	Allow
Other internal access	SITEA_BACKUP	SITEA_INTERNAL_NETWORKS	Any	Block + Log

Visitor separation	SITEA_BACKUP	SITEA_VISITOR	Any	Block + Log
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5.6 Printer VLAN Controls

Control	Source	Destination	Service	Action
DNS	SITEA_PRINTER	192.168.130.1	TCP/UDP 53	Allow
Gateway ICMP	SITEA_PRINTER	192.168.130.1	ICMP	Allow
Printer internal restriction	SITEA_PRINTER	SITEA_INTERNAL_NETWORKS	Any	Block + Log
Visitor separation	SITEA_PRINTER	SITEA_VISITOR	Any	Block + Log
web/update access	SITEA_PRINTER	Any	TCP 80/443	Allow

5.7 Management VLAN Controls

The Site A Management VLAN is used for administrative access to important network devices and services. Access is limited to the authorised management workstation rather than allowing the whole VLAN unrestricted access.

Control	Source	Destination	Service	Action
Management DNS	192.168.199.71	192.168.199.1	TCP/UDP 53	Allow
Gateway ICMP	192.168.199.71	192.168.199.1	ICMP	Allow
Admin to internal networks	192.168.199.71	SITEA_INTERNAL_NETWORKS	Any	Allow
Management web access	192.168.199.71	Any	TCP 80/443	Allow

5.8 Server VLAN Controls

The Site A Server VLAN allows the services needed for communication with the Backup VLAN and selected HQ server systems. Other internal and Visitor network access is blocked unless it is specifically required.

Control	Source	Destination	Service	Action
DNS	SITEA_SERVER	192.168.190.1	TCP/UDP 53	Allow
Gateway ICMP	SITEA_SERVER	192.168.190.1	ICMP	Allow
Server to Backup	SITEA_SERVER	SITEA_BACKUP	Prototype Any	Allow
Site A to HQ Server network	SITEA_SERVER	192.168.90.0/24	Approved VPN traffic	Allow
Other internal access	SITEA_SERVER	SITEA_INTERNAL_NETWORKS	Any	Block + Log
Visitor separation	SITEA_SERVER	SITEA_VISITOR	Any	Block + Log
Web access	SITEA_SERVER	Any	TCP 80/443	Allow

5.9 IPsec Interface Rules

The IPsec rules control what traffic is allowed through the site-to-site VPN after it reaches the firewall. Only the required traffic between the approved HQ and Site A networks is permitted. The complete IPsec firewall rules for both locations are covered in Section 6.2.

6. Inter-Site and VPN Firewall Rules

The Brisbane HQ and Mine Site A pfSense firewalls are used as the main endpoints for the IPsec VPN tunnel. The WAN firewall rules only allow the traffic needed to create and maintain the IKEv2/IPsec connection between both locations. Once the VPN traffic reaches the other firewall and is decrypted, the IPsec firewall rules are then used to control what traffic is actually allowed to pass between the HQ and Site A networks.

6.1 WAN VPN rules

Firewall	Remote peer	Local WAN	Required traffic	Action
HQ-FW WAN	172.16.101.2	172.16.100.2	UDP 500 / UDP 4500 / ESP	Allow
SITE-A-FW WAN	172.16.100.2	172.16.101.2	UDP 500 / UDP 4500 / ESP	Allow

6.2 IPsec Interface Rules

Firewall	Source	Destination	Service	Purpose
HQ-FW	192.168.110.0/24	192.168.10.0/24	ICMP	Allow Site A Employee to HQ Employee testing.
HQ-FW	192.168.190.0/24	192.168.90.60	UDP 514	Allow Site A firewall logging to HQ-MONITOR.
HQ-FW	192.168.190.0/24	192.168.90.0/24	ICMP	Allow Site A Server to HQ Server validation.
SITE-A-FW	192.168.10.0/24	192.168.110.0/24	ICMP	Allow HQ Employee to Site A Employee testing.
SITE-A-FW	192.168.90.0/24	192.168.190.0/24	ICMP	Allow HQ Server to Site A Server validation

These firewall rules work together with the Phase 2 selectors for the Employee networks, with the Brisbane HQ using 192.168.10.0/24 and Mine Site A using 192.168.110.0/24, with the server networks using 192.168.90.0/24 and 192.168.190.0/24. The Phase 2 selectors decide which networks can use the VPN, while the firewall rules decide what traffic is actually allowed between HQ and Mine Site A.

```

labadmin@srv-lnx-01:~$ sudo tcpdump -ni ens4 'esp or udp port 4500 or udp port 500'
[sudo: authenticate] Password:
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on ens4, link-type EN10MB (Ethernet), snapshot length 262144 bytes
05:38:29.071883 IP 172.16.100.2.500 > 172.16.101.2.500: isakmp: child_sa inf2
05:38:29.073510 IP 172.16.101.2.500 > 172.16.100.2.500: isakmp: child_sa inf2[I]
05:38:29.073629 IP 172.16.101.2.500 > 172.16.100.2.500: isakmp: child_sa inf2[IR]
05:38:29.074734 IP 172.16.100.2.500 > 172.16.101.2.500: isakmp: child_sa inf2[R]
05:38:30.111653 IP 172.16.100.2 > 172.16.101.2: ESP(spi=0xcb2aac49,seq=0x5), length 136
05:38:30.118601 IP 172.16.101.2 > 172.16.100.2: ESP(spi=0xcc46828d,seq=0x5), length 136
05:38:31.112854 IP 172.16.100.2 > 172.16.101.2: ESP(spi=0xcb2aac49,seq=0x6), length 136
05:38:31.117034 IP 172.16.101.2 > 172.16.100.2: ESP(spi=0xcc46828d,seq=0x6), length 136
05:38:32.114329 IP 172.16.100.2 > 172.16.101.2: ESP(spi=0xcb2aac49,seq=0x7), length 136
05:38:32.118241 IP 172.16.101.2 > 172.16.100.2: ESP(spi=0xcc46828d,seq=0x7), length 136
05:38:33.115606 IP 172.16.100.2 > 172.16.101.2: ESP(spi=0xcb2aac49,seq=0x8), length 136
05:38:33.120664 IP 172.16.101.2 > 172.16.100.2: ESP(spi=0xcc46828d,seq=0x8), length 136
05:38:39.072432 IP 172.16.100.2.500 > 172.16.101.2.500: isakmp: child_sa inf2
05:38:39.073953 IP 172.16.101.2.500 > 172.16.100.2.500: isakmp: child_sa inf2[I]
05:38:39.074060 IP 172.16.101.2.500 > 172.16.100.2.500: isakmp: child_sa inf2[IR]
05:38:39.075314 IP 172.16.100.2.500 > 172.16.101.2.500: isakmp: child_sa inf2[R]
05:38:49.073296 IP 172.16.100.2.500 > 172.16.101.2.500: isakmp: child_sa inf2
05:38:49.074804 IP 172.16.101.2.500 > 172.16.100.2.500: isakmp: child_sa inf2[I]
05:38:49.075223 IP 172.16.101.2.500 > 172.16.100.2.500: isakmp: child_sa inf2[IR]
05:38:49.076175 IP 172.16.100.2.500 > 172.16.101.2.500: isakmp: child_sa inf2[R]
^C
20 packets captured
20 packets received by filter
0 packets dropped by kernel
labadmin@srv-lnx-01:~$ _

```

Figure 4: HQ-ISP packet capture showing IKE and ESP-encrypted traffic between the HQ and Site A firewall WAN addresses.

7. Isolate Visitor Network

One of the main security controls I worked on was keeping the Visitor network separated from the internal PrimeCore networks. While the Visitor VLAN can still access services such as the internet, the firewall rule R04 blocks it from accessing the INTERNAL_NETWORKS alias.

This applies a restriction to visitor and contractor devices from reaching the internal areas of the Brisbane HQ, such as the Employee, Server and Management VLANs. Any event related to this rule, such as connection attempts, is logged and can be viewed using the implemented logging and monitoring solutions, such as Grafana and pfSense logging.

✗	Aug 27 15:59:06	VISITOR_VLAN	R04 - Block Visitor Access to Internal Networks (1787748853)	192.168.20.72	192.168.10.72	ICMP
✗	Aug 27 15:59:07	VISITOR_VLAN	R04 - Block Visitor Access to Internal Networks (1787748853)	192.168.20.72	192.168.10.72	ICMP

Figure 5: HQ R04 Visitor isolation rule showing blocked traffic from the Visitor VLAN to the internal network.



8. Employee Management Restrictions

Additionally, a security control that restricts employee devices from accessing the management VLAN was implemented. VLAN 99 uses the 192.168.99.0/24 network and is used for managing devices and services within the Brisbane HQ. I created separate pfSense firewall rules for the Employee LAN on VLAN 10 and Employee Wi-Fi on VLAN 11. The rules block both networks from accessing the Management VLAN, and logging is enabled so I can check any blocked connection attempts during testing.

Through testing that access from the Employee networks to the management VLAN was verified to be blocked, which keeps the management services separated from normal user devices.

9. Testing and Validation

Test	Expected result	Actual result	Status
HQ Visitor to HQ Employee	Blocked and logged	Blocked by R04 and logged	PASS
HQ Employee to HQ Management	Blocked and logged	Blocked during testing	PASS
HQ Visitor laptop to HQ-DC AD services	Only defined AD ports permitted	Domain join and GuestUser authentication succeeded.	PASS
HQ-MGMT-PC to Grafana	TCP 3000 permitted	Dashboard accessible	PASS
Site A Visitor DHCP	Receives 192.168.120.x address	DHCP successful	PASS
Site A Visitor to internal networks	Blocked	Employee/management tests blocked	PASS
Site A CCTV DHCP	Receives 192.168.141.x address	DHCP successful	PASS
Site A CCTV to Employee/Management	Blocked	Blocked as expected	PASS
Site A Backup to Server	Allowed	Connectivity successful	PASS
Site A Backup to Employee/Management	Blocked	Blocked as expected	PASS
Site A Video to own gateway	Allowed	Successful after gateway-specific ICMP rule	PASS
Site A Video to Management	Blocked	Blocked after rule correction	PASS
SITE-A-MGMT-PC to FW/SW/Employee	Allowed	All three tests were successful	PASS
Site A Server to Backup	Allowed	Successful	PASS
Site A Server to Employee/Management	Blocked	Blocked as expected	PASS

Site A Server to HQ-MONITOR	Allowed through VPN	Successful after HQ IPsec ICMP rule	PASS
Site A firewall logs to HQ	UDP 514 over VPN	Received and displayed in Grafana	PASS
Site A Printer endpoint	Dedicated endpoint validation	The printer endpoint successfully connected to VLAN 30 and received network access as expected	PASS

10. Security, Risk and Considerations

One of the main security risks identified was devices gaining access to networks they did not require. An example of this would be the visitor devices accessing internal PrimeCore networks or normal employee devices reaching the management VLAN. By implementing block rules, it reduces this risk while still allowing services needed for the user's role.

By enabling logging on the main firewall, block rules such as Visitor access to internal networks and Employee access to the management VLAN. Allowing for effective testing or implemented rules and traffic was being blocked.

When planning and implementing firewall rules, I considered what access each VLAN needed and what communication to avoid. Visitor, management and device networks were kept separated, while only the required services such as web, DNS and printing were allowed. I also considered the order of the firewall rules because more specific allow and block rules needed to be placed correctly before broader rules. Network and port aliases were used to make the configuration easier to understand and update as changes were made to the Brisbane HQ network.

11. Known Limitations

- Some of the Server and backup rules currently allow any traffic between the two networks. In a real network, these would be limited to only the ports and server addresses needed for the backup service.
- Some devices in the prototype, such as the management workstation, use static IP addresses. In a real deployment, this could be improved by using things like NAC, role-based access or centrally managed address reservations.
- The firewall rules would also need to be reviewed regularly to make sure old or unnecessary rules are removed and new services are only given the access they actually need.